

DATA MODERNIZATION & GIS SERVICE DISCOVERY

Marin County, CA | Health and Human Services

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Keywords:

Data Modernization, Data Governance, Metadata, GIS, Analytics Infrastructure

Summary:

To support the County's data modernization initiative, Atika partnered with the Epidemiology team to evaluate the current data ecosystem and develop recommendations for improving data governance, storage, and analytical workflows. By conducting stakeholder interviews, mapping data workflows, and documenting governance requirements, she identified key opportunities to centralize data management, standardize data classification, and improve metadata documentation through a SharePoint-based catalog. In addition, Atika led a county-wide GIS service discovery to assess departmental GIS capabilities and identify opportunities for greater collaboration and strategic alignment. This work provides a roadmap for future database modernization, secure data sharing, and automated data pipelines across the organization.

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Health and Human Services

Marin County, CA

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coding it forward >



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PROJECT OBJECTIVES

- Understand how the Epidemiology team stores, processes, and shares data.
- Identify opportunities to improve data governance, documentation, and efficiency.
- Develop a metadata catalog framework to support future data management.
- Evaluate options for centralized data storage.
- Conduct a county-wide GIS service discovery to understand GIS services and collaboration across departments



METHODS

Discovery

- Interviewed Epidemiology and GIS staff
- Observed current workflows
- Reviewed data governance documentation

Analysis

- Mapped the current data ecosystem
- Identified recurring pain points
- Compared current practices with data modernization principles

Design

- Developed a metadata catalog structure
- Designed a data classification framework
- Began evaluating centralized database options

EPIDEMIOLGY DATA GOVERNANCE

Understanding data storage, governance, and metadata practices

KEY FINDINGS

Finding

Impact

Current Workflow

Distributed storage

Security Concern, Inefficiency

Network latency

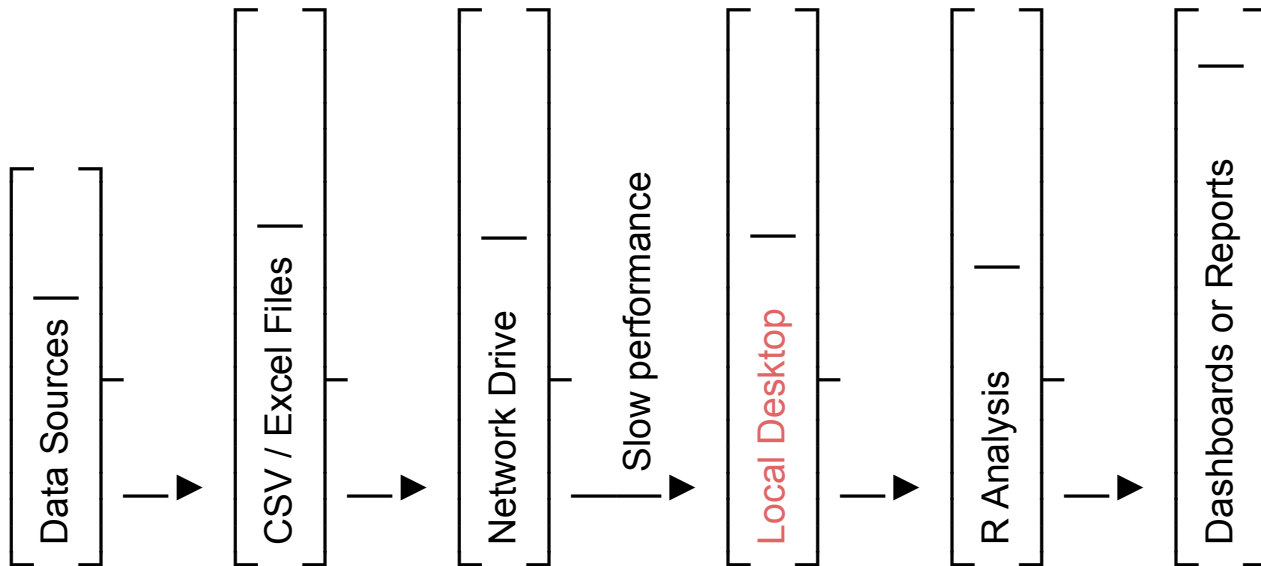
Local copies

Lack of Governance

Difficult to locate datasets, time consuming, difficult to train new staff

Manual workflows

Time-consuming, Risky



DATABASE EVALUATION CRITERIA

Feature	PostgreSQL	AWS (RDS/S3)	Azure SQL	Snowflake	DuckDB
Open source	✓	✗	✗	✗	✓
Centralized storage	✓	✓	✓	✓	Limited
Multi-user	✓	✓	✓	✓	✗
Role-based permissions	✓	✓	✓	✓	✗
R integration	Excellent	Excellent	Excellent	Excellent	Excellent
API support	Yes	Yes	Yes	Yes	Limited
Metadata support	Yes	Yes	Yes	Yes	Basic
Cloud-ready	Self-host or managed	Native	Native	Native	No

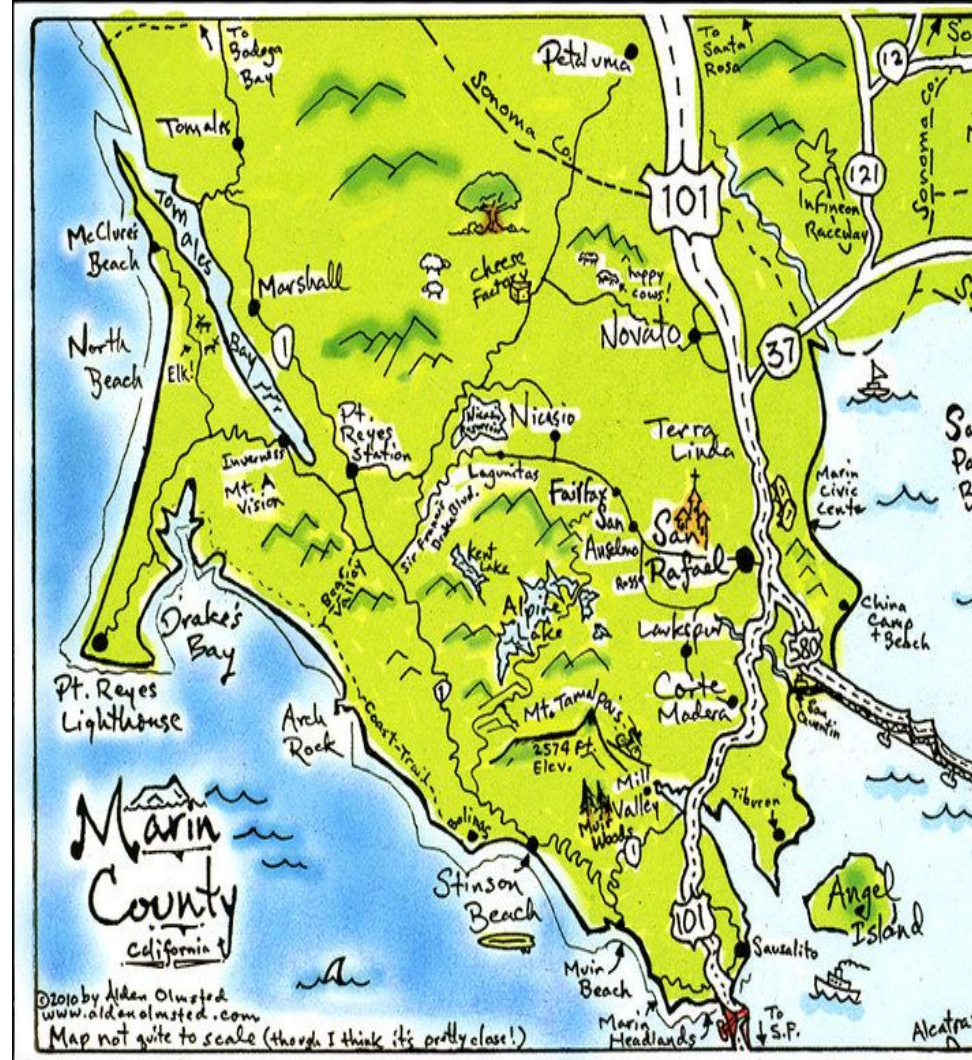
COUNTY-WIDE GIS SERVICE DISCOVERY

Interviews across OEM, CDA, DPW, Parks, and IST

KEY GIS FINDINGS

- **Mature, centralized infrastructure:** IST's Esri enterprise license (~400 users) now hosts GIS for all five departments interviewed.
- **Ownership and metadata are inconsistent:** Legacy dataset ownership is unclear for roughly half of countywide layers; documentation quality varies widely.
- **GIS is seen as “map-making,” not a platform:** Departments request one-off maps instead of defining data needs, leaving many requests only partly answerable.
- **Public-facing services are fragmented:** Accessibility rules and overlapping sources (MarinMap, state, federal) leave residents unsure which map is authoritative.
- **Staffing is thin relative to demand:** 3–4 GIS/data staff at IST plus 1–3 per department support roughly 400 licensed users countywide.

- **Assign clear data ownership:** Formalize who owns, updates, and approves release for each dataset — in Epidemiology and across GIS-holding departments.
- **Shift from map requests to data requests:** Route one-off map asks through a business-needs process that surfaces the underlying data gap.
- **Coordinate GIS investment countywide:** Use this discovery as a baseline for aligning standards, staffing, and shared services across IST, CDA, DPW, Parks, and OEM.



REFLECTIONS

- Centralized licensing and shared servers didn't fix data governance; ownership, metadata, and documentation gaps persisted regardless of the platform.
- The Epidemiology team and several county departments rely on one or two people's institutional memory for onboarding, cleaning, and workflows.
- Departments frequently ask for a map when the real gap is missing, undocumented, or ungoverned data underneath it.
- Even when standardization and governance is lacking, cross-department data sharing works well when people know each other personally/
- A metadata catalog and tier classification system are only useful if paired with documentation staff can realistically follow.

Today	Future
Data scattered across network drives and desktops	Data stored in a centralized database with documented data catalog
No standardized classification	Classification and sensitivity informed by Tier 1/2/3 + PSC scoring on every dataset
Onboarding, QA processes dependent on one individual	Documented QA and Onboarding process with assigned roles
Inconsistent naming conventions, functions, analysis process	R packages for the entire HHS department, standardized naming format
Unclear ownership across legacy GIS datasets	Defined ownership and metadata by department
Open Office documents, spreadsheets, PDFs	Standardized metadata, data formats, and data types

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